

ID: 6ae1360d

A circle has a radius of **2.1** inches. The area of the circle is $b\pi$ square inches, where b is a constant. What is the value of b ?

ID: 9ec76b54

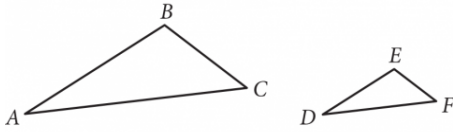
A right triangle has legs with lengths of **28** centimeters and **20** centimeters. What is the length of this triangle's hypotenuse, in centimeters?

- A. $8\sqrt{6}$
- B. $4\sqrt{74}$
- C. 48
- D. 1,184

ID: f1c1e971

The measure of angle R is $\frac{2\pi}{3}$ radians. The measure of angle T is $\frac{5\pi}{12}$ radians greater than the measure of angle R . What is the measure of angle T , in degrees?

- A. 75
- B. 120
- C. 195
- D. 390

ID: 1c3d613c

Note: Figures not drawn to scale.

Triangle ABC and triangle DEF are shown. The relationship between the side lengths of the two triangles is such that $\frac{AB}{DE} = \frac{BC}{EF} = \frac{AC}{DF} = 3$.

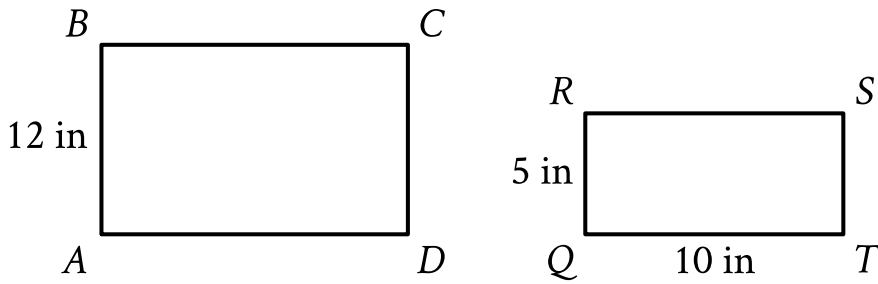
If the measure of angle BAC is 20° , what is the measure, in degrees, of angle EDF ? (Disregard the degree symbol when gridding your answer.)

ID: 2d521ca9

The measure of angle Z is 60° . What is the measure, in radians, of angle Z ?

- A. $\frac{1}{6}\pi$
- B. $\frac{1}{3}\pi$
- C. $\frac{2}{3}\pi$
- D. 1π

ID: e9c5bf2



Note: Figure not drawn to scale.

Rectangles $ABCD$ and $QRST$ shown are similar, where A , B , C , and D correspond to Q , R , S , and T , respectively. What is the length, in inches (**in**), of $\overline{AD}$?

- A. 60
- B. 24
- C. 17
- D. 10

ID: 08b7a3f5

A triangular prism has a height of **8 centimeters (cm)** and a volume of **216 cm³**. What is the area, **in cm²**, of the base of the prism? (The volume of a triangular prism is equal to **Bh** , where **B** is the area of the base and **h** is the height of the prism.)

ID: 7a8ad237

Triangles ABC and DEF are congruent, where A corresponds to D , and B and E are right angles. The measure of angle A is 69° . What is the measure, in degrees, of angle F ?

ID: 856372ca

In the xy -plane, a circle with radius 5 has center $(-8, 6)$. Which of the following is an equation of the circle?

A. $(x - 8)^2 + (y + 6)^2 = 25$

B. $(x + 8)^2 + (y - 6)^2 = 25$

C. $(x - 8)^2 + (y + 6)^2 = 5$

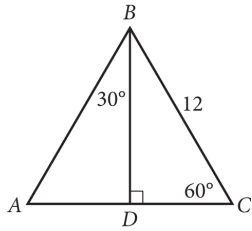
D. $(x + 8)^2 + (y - 6)^2 = 5$

ID: 9c0a0eca

A triangle has a base length of **10** centimeters and a corresponding height of **70** centimeters. What is the area, in square centimeters, of the triangle?

- A. **700**
- B. **350**
- C. **175**
- D. **80**

ID: bf8d843e



In $\triangle ABC$ above, what is the length of $\overline{AD}$?

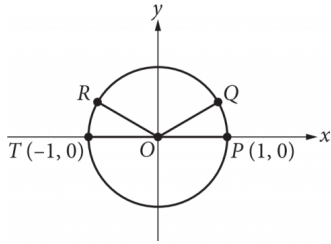
- A. 4
- B. 6
- C. $6\sqrt{2}$
- D. $6\sqrt{3}$

ID: aef4fd8a

The length of each side of a square is **94** centimeters (cm). Which expression gives the area, in **cm²**, of the square?

- A. **$2 \cdot 94$**
- B. **$2 \cdot 94 \cdot 94$**
- C. **$4 \cdot 94$**
- D. **$94 \cdot 94$**

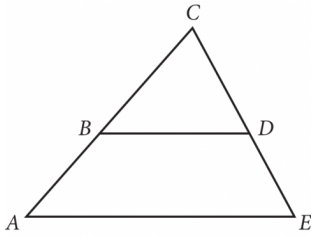
ID: 95ba2d09



In the xy -plane above, points P , Q , R , and T lie on the circle with center O . The degree measures of angles $\angle POQ$ and $\angle ROT$ are each 30° . What is the radian measure of angle $\angle QOR$?

- A. $\frac{5}{6}\pi$
- B. $\frac{3}{4}\pi$
- C. $\frac{2}{3}\pi$
- D. $\frac{1}{3}\pi$

ID: 6dd463ca

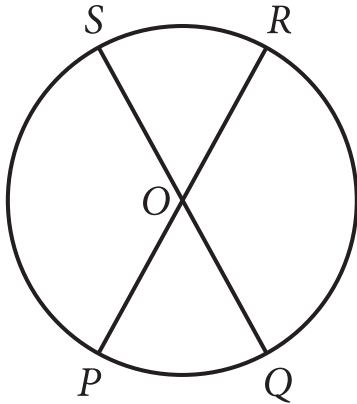


Note: Figure not drawn to scale.

In the figure above, segments AE and BD are parallel. If angle BDC measures 58° and angle ACE measures 62° , what is the measure of angle CAE ?

- A. 58°
- B. 60°
- C. 62°
- D. 120°

ID: 0815a5af

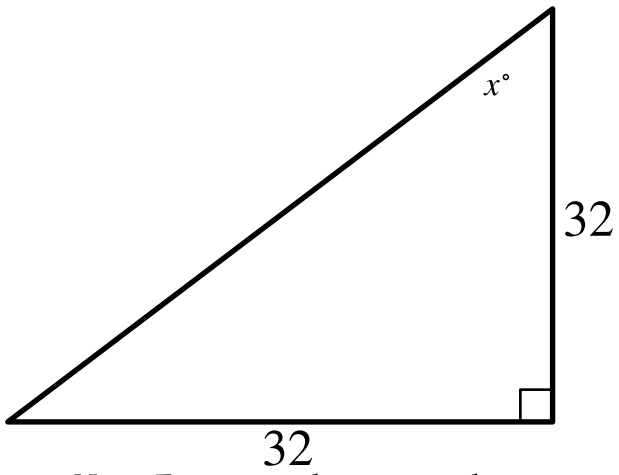


Note: Figure not drawn to scale.

The circle shown has center O , circumference 144π , and diameters $\overline{PR}$ and $\overline{QS}$. The length of arc PS is twice the length of arc PQ . What is the length of arc QR ?

- A. 24π
- B. 48π
- C. 72π
- D. 96π

ID: a71617d3



Note: Figure not drawn to scale.

In the triangle shown, what is the value of x ?

ID: a5aee181

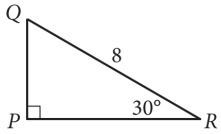
The length of a rectangle's diagonal is $5\sqrt{17}$, and the length of the rectangle's shorter side is 5. What is the length of the rectangle's longer side?

- A. $\sqrt{17}$
- B. 20
- C. $15\sqrt{2}$
- D. 400

ID: ae33e0a1

A right circular cylinder has a base diameter of **22** centimeters and a height of **6** centimeters. What is the volume, in cubic centimeters, of the cylinder?

- A. 132π
- B. 264π
- C. 726π
- D. $2,904\pi$

ID: 13d9a1c3

In the right triangle shown above, what is the length of $\overline{PQ}$?

ID: 4ff7b652

Right triangles LMN and PQR are similar, where L and M correspond to P and Q , respectively. Angle M has a measure of 53° . What is the measure of angle Q ?

- A. 37°
- B. 53°
- C. 127°
- D. 143°

ID: cecbdeba

A right circular cylinder has a volume of **432** cubic centimeters. The area of the base of the cylinder is **24** square centimeters. What is the height, in centimeters, of the cylinder?

- A. 18
- B. 24
- C. 216
- D. 10,368

ID: a2e76b60

A cylindrical can containing pieces of fruit is filled to the top with syrup before being sealed. The base of the can has an area of 75 cm^2 , and the height of the can is 10 cm. If 110 cm^3 of syrup is needed to fill the can to the top, which of the following is closest to the total volume of the pieces of fruit in the can?

- A. 7.5 cm^3
- B. 185 cm^3
- C. 640 cm^3
- D. 750 cm^3

ID: 468613c0

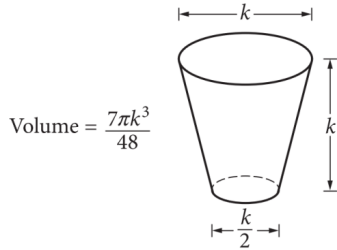
A triangle has a base length of **56** centimeters and a height of **112** centimeters. What is the area, in square centimeters, of the triangle?

- A. **168**
- B. **1,568**
- C. **3,136**
- D. **6,272**

ID: 95ca2683

In triangle ABC , the measure of angle A is 54° , the measure of angle B is 90° , and the measure of angle C is $\left(\frac{k}{2}\right)^\circ$. What is the value of k ?

- A. 36
- B. 45
- C. 72
- D. 108

ID: 37dde49f

The glass pictured above can hold a maximum volume of 473 cubic centimeters, which is approximately 16 fluid ounces. What is the value of k , in centimeters?

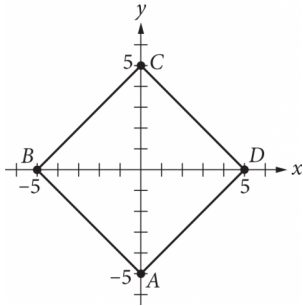
- A. 2.52
- B. 7.67
- C. 7.79
- D. 10.11

ID: 82c8325f

A circle in the xy -plane has its center at $(-4, 5)$ and the point $(-8, 8)$ lies on the circle. Which equation represents this circle?

- A. $(x - 4)^2 + (y + 5)^2 = 5$
- B. $(x + 4)^2 + (y - 5)^2 = 5$
- C. $(x - 4)^2 + (y + 5)^2 = 25$
- D. $(x + 4)^2 + (y - 5)^2 = 25$

ID: cf53cb56



In the xy -plane shown, square $ABCD$ has its diagonals on the x - and y -axes. What is the area, in square units, of the square?

- A. 20
- B. 25
- C. 50
- D. 100

ID: b96ff36e

In the xy -plane, the graph of the equation $(x - 3)^2 + (y - 5)^2 = 9$ is a circle. The point $(6, c)$, where c is a constant, lies on this circle. What is the value of c ?

ID: ee540927

$$x^2 + 58x + y^2 = 0$$

In the xy -plane, the graph of the given equation is a circle. What are the coordinates (x, y) of the center of the circle?

- A. $(0, 29)$
- B. $(0, -29)$
- C. $(29, 0)$
- D. $(-29, 0)$

ID: a0cacec1

An angle has a measure of $\frac{16\pi}{15}$ radians. What is the measure of the angle, in degrees?

ID: 74d8b897

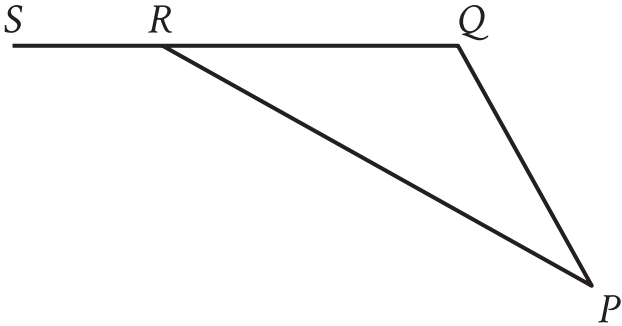
An angle has a measure of $\frac{9\pi}{20}$ radians. What is the measure of the angle in degrees?

ID: fc8aa563

What is the center of the circle in the xy -plane defined by the equation $(x - 1)^2 + (y + 7)^2 = 1$?

- A. $(-1, -7)$
- B. $(-1, 7)$
- C. $(1, -7)$
- D. $(1, 7)$

ID: 014edcb7



Note: Figure not drawn to scale.

In triangle PQR , $\overline{QR}$ is extended to point S . The measure of $\angle PQR$ is 132° , and the measure of $\angle PRS$ is 163° . What is the measure of $\angle QPR$?

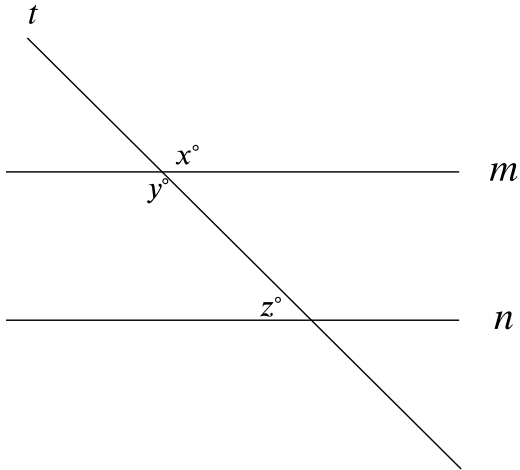
- A. 48°
- B. 31°
- C. 24°
- D. 17°

ID: 7700d098

One leg of a right triangle has a length of **43.2** millimeters. The hypotenuse of the triangle has a length of **196.8** millimeters. What is the length of the other leg of the triangle, in millimeters?

- A. **43.2**
- B. **120**
- C. **192**
- D. **201.5**

ID: 2adbf1b1

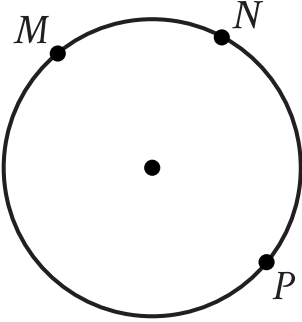


Note: Figure not drawn to scale.

In the figure, lines m and n are parallel. If $x = 6k + 13$ and $y = 8k - 29$, what is the value of z ?

- A. 3
- B. 21
- C. 41
- D. 139

ID: 800e71b8



Points M , N , and P lie on the circle shown. On this circle, minor arc MN has a length of **39** centimeters and major arc MPN has a length of **195** centimeters. What is the circumference, in centimeters, of the circle shown?

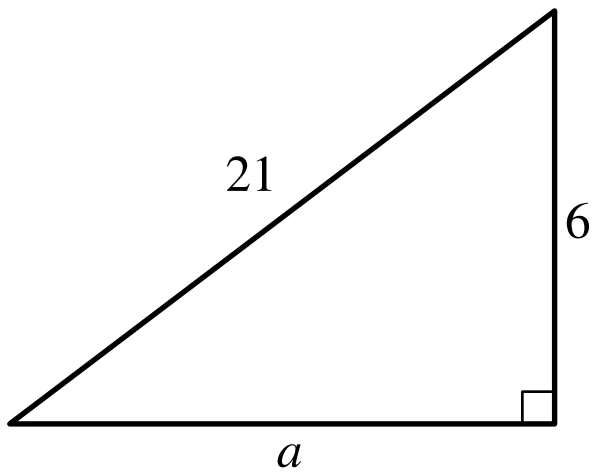
- A. **39**
- B. **156**
- C. **195**
- D. **234**

ID: 901e3285

In triangle ABC , the measure of angle A is 50° . If triangle ABC is isosceles, which of the following is NOT a possible measure of angle B ?

- A. 50°
- B. 65°
- C. 80°
- D. 100°

ID: de550be0



Note: Figure not drawn to scale.

For the triangle shown, which expression represents the value of a ?

- A. $\sqrt{21^2 - 6^2}$
- B. $21^2 - 6^2$
- C. $\sqrt{21 - 6}$
- D. $21 - 6$

ID: c7bed21d

Quadrilateral $P'Q'R'S'$ is similar to quadrilateral $PQRS$, where P , Q , R , and S correspond to P' , Q' , R' , and S' , respectively. The measure of angle P is 30° , the measure of angle Q is 50° , and the measure of angle R is 70° . The length of each side of $P'Q'R'S'$ is **3** times the length of each corresponding side of $PQRS$. What is the measure of angle P' ?

- A. 10°
- B. 30°
- C. 40°
- D. 90°

ID: 8e7689e0

The number of radians in a 720-degree angle can be written as $a\pi$, where a is a constant. What is the value of a ?

ID: 2085e10e

In triangle DEF , the measure of angle D is 47° and the measure of angle E is 97° . In triangle RST , the measure of angle R is 47° and the measure of angle S is 97° . Which of the following additional pieces of information is needed to determine whether triangle DEF is similar to triangle RST ?

- A. The measure of angle F
- B. The measure of angle T
- C. The measure of angle F and the measure of angle T
- D. No additional information is needed.

ID: fd8745fc

In triangle JKL , the measures of $\angle K$ and $\angle L$ are each 48° . What is the measure of $\angle J$, in degrees?
(Disregard the degree symbol when entering your answer.)

ID: d3fe472f

Triangle ABC is similar to triangle XYZ , such that A , B , and C correspond to X , Y , and Z respectively. The length of each side of triangle XYZ is **2** times the length of its corresponding side in triangle ABC . The measure of side AB is **16**. What is the measure of side XY ?

- A. **14**
- B. **16**
- C. **18**
- D. **32**

ID: aa2d36fe

A circle in the xy -plane has the equation $(x - 13)^2 + (y - k)^2 = 64$. Which of the following gives the center of the circle and its radius?

- A. The center is at $(13, k)$ and the radius is 8.
- B. The center is at $(k, 13)$ and the radius is 8.
- C. The center is at $(k, 13)$ and the radius is 64.
- D. The center is at $(13, k)$ and the radius is 64.

ID: 98c12e38

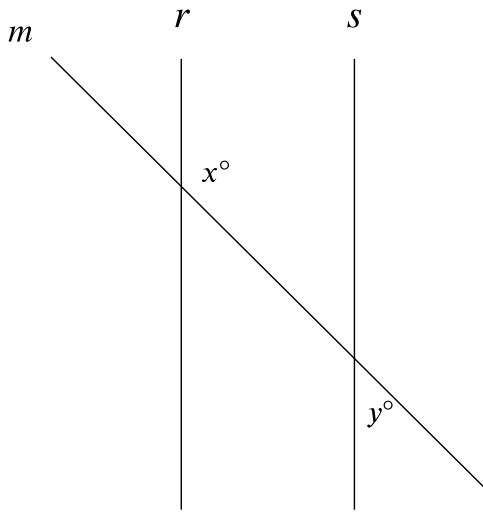
Circle K has a radius of 4 millimeters (mm). Circle L has an area of $100\pi \text{ mm}^2$. What is the total area, in mm^2 , of circles K and L ?

- A. 14π
- B. 28π
- C. 56π
- D. 116π

ID: 38517165

A circle has a circumference of 31π centimeters. What is the diameter, in centimeters, of the circle?

ID: a4c05a1b



Note: Figure not drawn to scale.

In the figure shown, lines r and s are parallel, and line m intersects both lines. If $y < 65$, which of the following must be true?

- A. $x < 115$
- B. $x > 115$
- C. $x + y < 180$
- D. $x + y > 180$

ID: e0874bc2

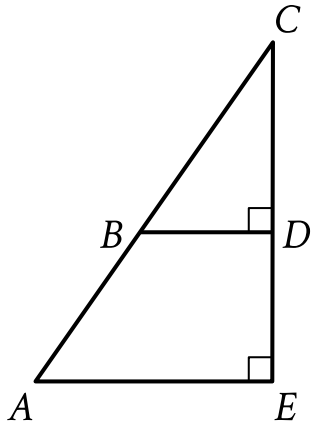
The table gives the perimeters of similar triangles TUV and XYZ , where $\overline{TU}$ corresponds to $\overline{XY}$. The length of $\overline{TU}$ is 18.

	Perimeter
Triangle TUV	37
Triangle XYZ	333

What is the length of $\overline{XY}$?

- A. 2
- B. 18
- C. 55
- D. 162

ID: 2f7c92ad



Note: Figure not drawn to scale.

In the figure shown, triangle CAE is similar to triangle CBD . The measure of angle CBD is 57° , and $AE = 26(BD)$. What is the measure of angle CAE ?

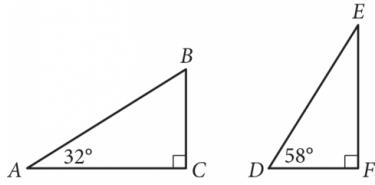
- A. $(26 \cdot 57)^\circ$
- B. $(26 + 57)^\circ$
- C. 57°
- D. 26°

ID: a2659088

A right circular cylinder has a height of **8 meters (m)** and a base with a radius of **12 m**. What is the volume, **in m³**, of the cylinder?

- A. 8π
- B. 20π
- C. 768π
- D. $1,152\pi$

ID: 933fee1a



Triangles ABC and DEF are shown above. Which of the

following is equal to the ratio $\frac{BC}{AB}$?

A. $\frac{DE}{DF}$

B. $\frac{DF}{DE}$

C. $\frac{DF}{EF}$

D. $\frac{EF}{DE}$

ID: d7a8aa9c

Each side of equilateral triangle S is multiplied by a scale factor of k to create equilateral triangle T. The length of each side of triangle T is greater than the length of each side of triangle S. Which of the following could be the value of k ?

- A. $\frac{29}{28}$
- B. 1
- C. $\frac{28}{29}$
- D. 0

ID: 5a7e3b46

In $\triangle ABC$, $\angle B$ is a right angle and the length of $\overline{BC}$ is **136** millimeters. If $\cos A = \frac{3}{5}$, what is the length, in millimeters, of $\overline{AB}$?

- A. **34**
- B. **102**
- C. **136**
- D. **170**

ID: d621cffb

A sphere has a radius of $\frac{17}{5}$ feet. What is the volume, in cubic feet, of the sphere?

- A. $\frac{5\pi}{17}$
- B. $\frac{68\pi}{15}$
- C. $\frac{32\pi}{5}$
- D. $\frac{19,652\pi}{375}$

ID: 151eda3c

A manufacturing company produces two sizes of cylindrical containers that each have a height of 50 centimeters. The radius of container A is 16 centimeters, and the radius of container B is 25% longer than the radius of container A. What is the volume, in cubic centimeters, of container B?

- A. $16,000\pi$
- B. $20,000\pi$
- C. $25,000\pi$
- D. $31,250\pi$

ID: 03c6994f

Square A has side lengths that are **246** times the side lengths of square B. The area of square A is ***k*** times the area of square B. What is the value of ***k***?

- A. **60,516**
- B. **492**
- C. **246**
- D. **123**

ID: c0586eb5

A cylinder has a diameter of **8** inches and a height of **12** inches. What is the volume, in cubic inches, of the cylinder?

- A. 16π
- B. 96π
- C. 192π
- D. 768π

ID: e4b4e9ea

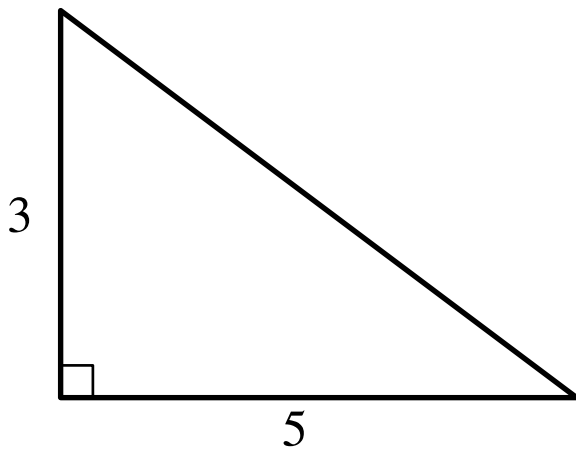
The length of the edge of the base of a right square prism is **6** units. The volume of the prism is **2,880** cubic units. What is the height, in units, of the prism?

- A. $4\sqrt{30}$
- B. **36**
- C. $24\sqrt{5}$
- D. **80**

ID: e336a1d2

A cube has an edge length of **41** inches. What is the volume, in cubic inches, of the cube?

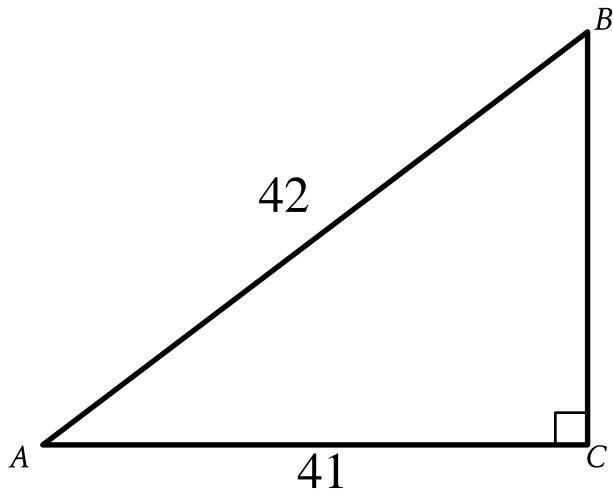
- A. **164**
- B. **1,681**
- C. **10,086**
- D. **68,921**

ID: a4ed5285

Note: Figure not drawn to scale.

The figure shows the lengths, in inches, of two sides of a right triangle. What is the area of the triangle, in square inches?

ID: 2bddbc1b



Note: Figure not drawn to scale.

What is the value of $\cos A$ in the triangle shown?

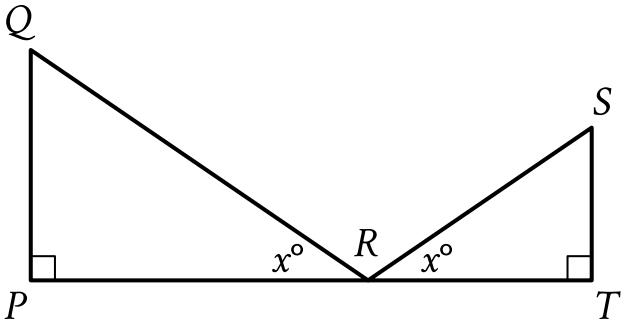
- A. $\frac{42}{41}$
- B. $\frac{41}{42}$
- C. $\frac{1}{42}$
- D. $\frac{1}{41}$

ID: 055aafe7

Triangle ABC is similar to triangle XYZ , where A , B , and C correspond to X , Y , and Z , respectively. In triangle ABC , the length of $\overline{AB}$ is 170 and the length of $\overline{BC}$ is 850. In triangle XYZ , the length of $\overline{YZ}$ is 60. What is the length of $\overline{XY}$?

- A. 204
- B. 182
- C. 60
- D. 12

ID: 51f26ce8



Note: Figure not drawn to scale.

$\triangle QPR$ is similar to $\triangle STR$. The lengths represented by $\overline{ST}$, $\overline{QP}$, $\overline{PR}$, and $\overline{QR}$ in the figure are 14, 15, 20, and 25, respectively. What is the length of $\overline{SR}$?

- A. $\frac{350}{15}$
- B. $\frac{350}{20}$
- C. $\frac{210}{20}$
- D. $\frac{210}{25}$

ID: 94364a79

Two nearby trees are perpendicular to the ground, which is flat. One of these trees is **10** feet tall and has a shadow that is **5** feet long. At the same time, the shadow of the other tree is **2** feet long. How tall, in feet, is the other tree?

- A. **3**
- B. **4**
- C. **8**
- D. **27**

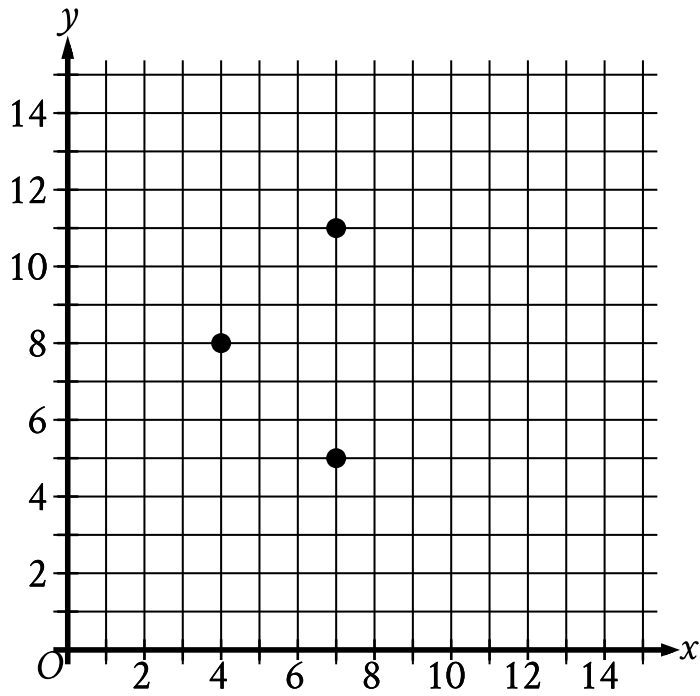
ID: 3b931fb0

A right circular cylinder has a volume of **377** cubic centimeters. The area of the base of the cylinder is **13** square centimeters. What is the height, in centimeters, of the cylinder?

ID: ec5d4823

What is the volume, in cubic centimeters, of a right rectangular prism that has a length of 4 centimeters, a width of 9 centimeters, and a height of 10 centimeters?

ID: 096c7ef5



The three points shown define a circle. The circumference of this circle is $k\pi$, where k is a constant. What is the value of k ?

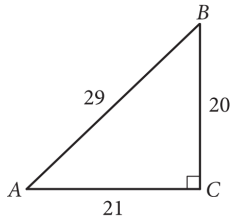
- A. 3
- B. 6
- C. 7
- D. 9

ID: e86f0651

A circle has a radius of **43** meters. What is the area, in square meters, of the circle?

- A. $\frac{43\pi}{2}$
- B. 43π
- C. 86π
- D. $1,849\pi$

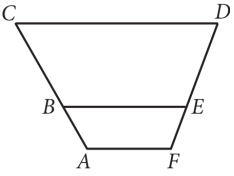
ID: 902dc959



In the figure above, what is the value of $\tan(A)$?

- A. $\frac{20}{29}$
- B. $\frac{21}{29}$
- C. $\frac{20}{21}$
- D. $\frac{21}{20}$

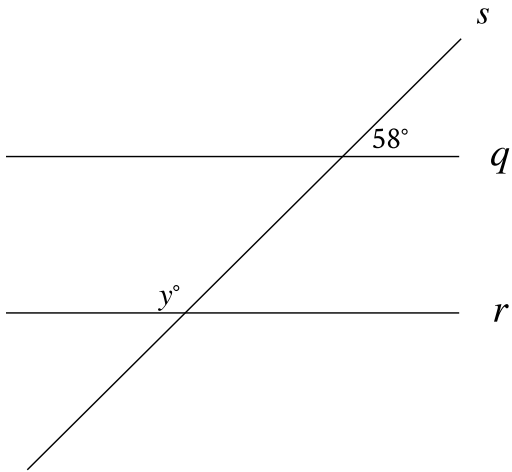
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In the figure above, $\overline{AF}$, $\overline{BE}$, and $\overline{CD}$ are parallel. Points B and E lie on $\overline{AC}$ and $\overline{FD}$, respectively. If $AB = 9$, $BC = 18.5$, and $FE = 8.5$, what is the length of $\overline{ED}$, to the nearest tenth?

- A. 16.8
- B. 17.5
- C. 18.4
- D. 19.6

ID: 686b5212



Note: Figure not drawn to scale.

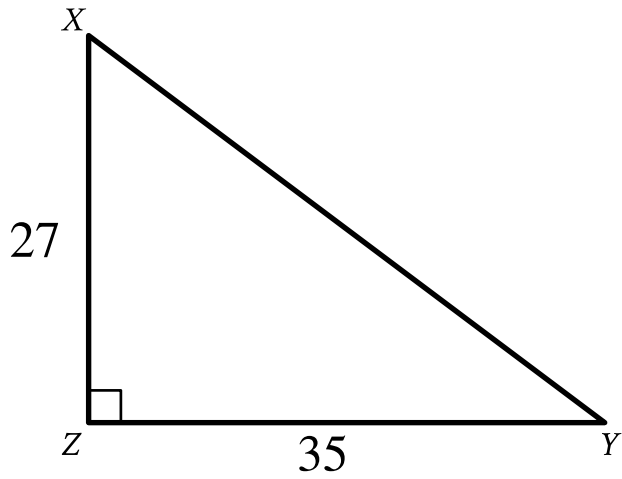
In the figure, line q is parallel to line r , and both lines are intersected by line s . If $y = 2x + 8$, what is the value of x ?

ID: 1f0b582e

Square X has a side length of **12** centimeters. The perimeter of square Y is **2** times the perimeter of square X. What is the length, in centimeters, of one side of square Y?

- A. **6**
- B. **10**
- C. **14**
- D. **24**

ID: 659cb706



Note: Figure not drawn to scale.

Triangle XYZ shown is a right triangle. Which of the following has the same value as $\sin X$?

- A. $\tan X$
- B. $\tan Y$
- C. $\cos X$
- D. $\cos Y$

ID: 5afbdc8e

What is the length of one side of a square that has the same area as a circle with radius 2 ?

- A. 2
- B. $\sqrt{2\pi}$
- C. $2\sqrt{\pi}$
- D. 2π

ID: 33e29881

In right triangle RST , the sum of the measures of angle R and angle S is 90 degrees. The value of $\sin(R)$ is $\frac{\sqrt{15}}{4}$. What is the value of $\cos(S)$?

- A. $\frac{\sqrt{15}}{15}$
- B. $\frac{\sqrt{15}}{4}$
- C. $\frac{4\sqrt{15}}{15}$
- D. $\sqrt{15}$

ID: f67e4efc

A right circular cylinder has a volume of 45π . If the height of the cylinder is 5, what is the radius of the cylinder?

- A. 3
- B. 4.5
- C. 9
- D. 40

ID: bcb66188

Triangle FGH is similar to triangle JKL , where angle F corresponds to angle J and angles G and K are right angles. If $\sin(F) = \frac{308}{317}$, what is the value of $\sin(J)$?

- A. $\frac{75}{317}$
- B. $\frac{308}{317}$
- C. $\frac{317}{308}$
- D. $\frac{317}{75}$

ID: 4b7bb316

The length of each edge of a box is **29** inches. Each side of the box is in the shape of a square. The box does not have a lid. What is the exterior surface area, in square inches, of this box without a lid?